

AI-Powered Personalized News Aggregator

1. Project Idea

I propose building an **AI-powered personalized news aggregation system** that automatically collects, processes, and delivers the most relevant AI-related news to users.

The system will:

- Collect data from multiple sources (YouTube, blogs, RSS feeds)
- Store it in a structured database
- Use AI (LLMs) to summarize and analyze content
- Rank news based on user interests
- Deliver a **daily personalized email digest**

This project simulates a **real-world intelligent assistant** that saves time and improves information consumption.

2. Project Objectives

The main goals of this project are:

- Apply **Data Science and AI concepts** in a real system
- Build an **end-to-end pipeline** (data collection → processing → output)
- Use **Large Language Models (LLMs)** in a practical application
- Design a **scalable and automated system**
- Demonstrate strong skills in **backend development and system design**

3. System Overview (How It Works)

The system works as a pipeline:

1. **Data Collection**
 - Extract news from:
 - YouTube (via RSS + transcripts)
 - AI blogs (OpenAI, Anthropic)
 - Other RSS sources
2. **Data Storage**
 - Store raw data in a **PostgreSQL database**
3. **AI Processing**

- Use an LLM to:
 - Summarize articles
 - Extract key insights
- 4. **Personalization & Ranking**
 - Rank content based on user interests using an AI agent
- 5. **Output Generation**
 - Generate a **daily email digest**
 - Send it automatically to the user
- 6. **Automation**
 - Run the system every 24 hours using a scheduled job (cron)

4. AI & Data Science Concepts Used

This project covers important topics from the diploma:

- **Natural Language Processing (NLP)**
- **Large Language Models (LLMs)**
- **Prompt Engineering**
- **Information Retrieval**
- **Data Pipelines**
- **Recommendation Systems (basic personalization)**
- **Data Storage & Management (SQL)**

5. Technologies Used

- **Python** (main programming language)
- **PostgreSQL** (database)
- **SQLAlchemy** (ORM)
- **OpenAI API** (LLM processing)
- **Docker** (environment setup)
- **Render** (cloud deployment)
- **SMTP (Gmail)** (email sending)

6. Innovation & Strength of the Project

This project is strong for graduation because:

- It is **end-to-end** (not just a model, but a full system)
- It uses **modern AI technologies (LLMs)**

- It solves a **real-world problem** (information overload)
- It includes **automation and deployment** (industry-level skill)
- It demonstrates **system design, not only coding**

7. Possible Extensions (Advanced Level)

If time allows, I can enhance the project with:

- Web dashboard (FastAPI + frontend)
- Multi-user support
- Feedback learning system (user clicks improve recommendations)
- RAG (Retrieval-Augmented Generation)
- Chat interface for querying news

8. Expected Deliverables

- Fully working system (code + deployment)
- Database with real data
- Automated daily email system
- Documentation (README + architecture diagram)
- Demo (live or recorded)